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Strengthening Intervention Approaches: The Influence of Multilateral Technologies on Field Development

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Abstract

Effective well completions are fundamental to optimizing well performance. Innovative planning and design play a key role in improving this process, enabling the elimination of observation wells and reducing the abandonment of pilot holes. Despite advancements such as lateral re-entry windows that provide access to each drilled lateral, interventions remain necessary. Therefore, a seamless completion design is essential to facilitate interventions and avoid operational challenges.

Lateral access windows significantly enhance the efficiency and effectiveness of well completions when thoughtfully integrated. Multilateral wells should be engineered to allow smooth interventions while maximizing well performance. The use of permanent downhole gauges in these wells enables continuous monitoring of pressure and temperature, delivering valuable data for timely operational decisions. Incorporating additional accessories further improves system reliability and provides contingency options in case of accessory failures. A well-considered design that addresses operational challenges is critical to maintaining system integrity and performance.

Seamless operations during interventions in multilateral wells are vital to minimizing risks and ensuring sustained well functionality. Periodic logging and monitoring activities, including intervention and production logging, are typically conducted in the vertical wellbore, which acts as the observation lateral. The lateral access window effectively supports these operations by enabling interventions in both the mother bore and other laterals. Field data has demonstrated the effectiveness of lateral access applications in advancing well development, with multiple wells successfully undergoing interventions for logging and treatment in both vertical and horizontal laterals.

To facilitate interventions in horizontal laterals, slickline can be used to retrieve an isolation sleeve and install a tubing exit whipstock (TEW) for lateral re-entry. Following this, coiled tubing or wireline equipped with a tractor is deployed through the window into the lateral to perform intervention operations such as production logging. After completing the intervention, the TEW is removed, the isolation sleeve reinstalled, and production resumed.

This multilateral completion technology exemplifies a design that simplifies and enhances access to both vertical and horizontal laterals without requiring rig mobilization or complex retrieval processes. Its lateral access capabilities improve real-time monitoring and support operational decisions, while streamlining

interventions and enhancing overall well performance. The technology also offers a range of completion methods, each with distinct advantages, demonstrating the versatility and effectiveness of multilateral completions.

Introduction

Multilateral well technology has evolved significantly over recent decades, driven by the increasing complexity of reservoirs and the need to optimize field development economics. Early multilateral completions were limited in scope and effectiveness, often leaving valuable reservoir zones underdeveloped. However, advances in drilling and completion techniques have propelled multilateral wells to the forefront of modern reservoir management strategies, enabling greater value extraction from hydrocarbon assets. Advanced multilateral well design incorporates robust completion technology such as lateral access windows, and advanced monitoring systems such as permanent downhole gauges (PDGs). These features ensure reliable access for interventions, continuous real-time data collection, and dynamic adaptation to changing reservoir conditions. The integration of these technologies supports intelligent completion strategies, enhances integrity, and facilitates efficient reservoir management throughout the well's lifecycle. Despite these advantages, multilateral wells present unique challenges, particularly in well construction, junction reliability, and intervention access. Addressing these challenges requires careful planning, innovative design, and the use of specialized equipment to maintain operational flexibility and data integrity. This introduction sets the stage for a detailed discussion of multilateral well design, access methodology, and the transformative value these wells bring to field development (Sharma, 2022).

The Role of Multilateral Wells in Field Development

Multilateral well technology has undergone remarkable evolution over the past several decades, driven by the increasing complexity of reservoirs and the need to optimize field development economics. Early multilateral completions were relatively minimal, limiting their effectiveness and leaving many reservoir zones underdeveloped or untapped. However, recent technological advancements have propelled multilateral wells to the forefront of reservoir management strategies, enabling operators to unlock greater value from their assets.

The principal advantage of multilateral wells lies in their ability to significantly increase reservoir contact from a single wellbore. By drilling multiple lateral branches from one main bore, these wells can simultaneously access several reservoir zones, thereby maximizing hydrocarbon recovery while minimizing the number of surface locations required. This approach not only enhances production efficiency but also reduces drilling and completion costs, lowers the footprint of surface infrastructure, and mitigates environmental impacts-factors that are especially critical in offshore and environmentally sensitive areas.

Economically, multilateral wells offer substantial benefits by decreasing rig time and materials, reducing the need for multiple vertical wells, and simplifying surface and subsea infrastructure. This consolidation leads to lower capital and operational expenditures, accelerating time to first production and improving the overall project net present value (Almasri, 2024).

Well Construction Methods

The Technology Advancement of Multilaterals (TAML) system categorizes multilateral wells according to junction functionality and complexity, designated TAML Levels 1 through 6. This ranking system is used to determine the appropriate lateral design.

TAML levels are defined as follows:

1. TAML Level 1: An open hole lateral is drilled from an open hole main bore (Fig. 1). Level 1 multilaterals are drilled in fields with stable formations; however, flow control and lateral re-entry cannot be achieved.
2. TAML Level 2: The main bore is cased and cemented and the lateral is an open hole or drop liner without connection at the junction (Fig. 2). Flow control for each lateral can be achieved when an isolation packer is installed below each window. However, lateral access is not possible unless the entire completion is retrieved.
3. TAML Level 3: The main bore is cased and cemented and the lateral is cased but uncemented. The lateral wellbore is mechanically joined to the main bore, but the junction is not hydraulically sealed (Fig. 3). Level 3 construction is not common in the oil industry.
4. TAML Level 4: Both the main bore and lateral are cased and cemented and hydraulic integrity depends on the cement quality (Fig. 4).
5. TAML Level 5: The junction integrity is accomplished by the completion, and the junction can be cemented or uncemented (Fig. 5).

   Level 5 system is common in fields where gas migration at the junction is detected.
6. TAML Level 6: Two branches are drilled from the original wellbore, and both mechanical and pressure integrity are achieved using the casing to seal the junction (Fig. 6).

Figure 1—TAML Level 1.

Figure 2—TAML Level 2.

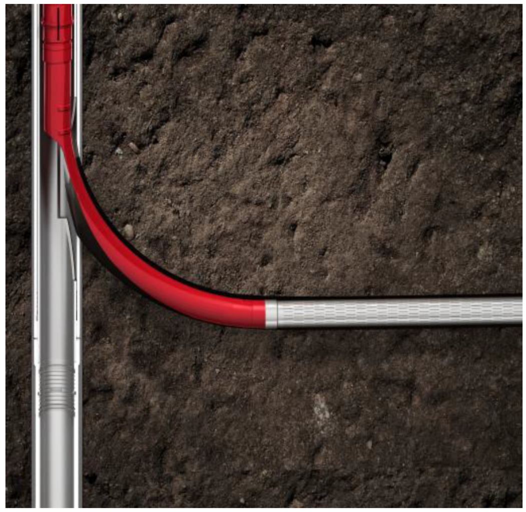

Figure 3—TAML Level 3.

Figure 4—TAML Level 4.

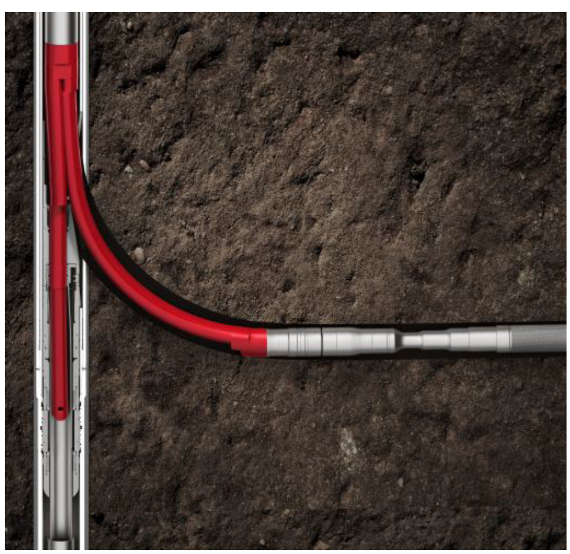

Figure 5—TAML Level 5.

Figure 6—TAML Level 6.

Completion Design Considerations for Multilateral Wells

The design of a multilateral well begins with the drilling phase, which is critical to ensuring long-term well integrity and operational reliability. Proper drilling practices establish a solid foundation that enables the well to sustain production over extended periods without compromising safety or performance. One of the most important design decisions involves selecting the appropriate junction type, which balances well integrity, operational complexity, and cost.

The choice of junction is guided by a thorough understanding of the wellbore conditions, reservoir characteristics, and the optimal sidetracking strategy. The simplest and most cost-effective junction is classified as TAML Level 2. This design features a cased and cemented main bore, while the lateral and junction remain open hole-uncased and uncemented-with the window cut through the cemented main bore. TAML Level 2 is suitable when the junction is located within the reservoir or when the formation at the junction is similar to the reservoir, allowing for stable wellbore conditions without cement support. However, this approach provides limited well integrity and no hydraulic isolation at the junction, which can pose risks in more challenging formations.

When well integrity concerns arise or conditions are less favorable, it becomes necessary to upgrade to a more robust junction design, such as TAML Level 4. At this level, both the main bore and the lateral are fully cased and cemented at the junction, providing enhanced mechanical support and protection against formation collapse or sand infiltration. Although TAML Level 4 junctions do not provide hydraulic isolation, they significantly improve wellbore stability and reduce the likelihood of costly workovers or remedial interventions (Al-Raml, 2023).

Beyond the junction design, the completion system must be fully compatible with the multilateral architecture throughout all phases-well construction, completion, and future interventions. Effective completion design ensures easy and reliable access to each lateral without jeopardizing the integrity of the overall well system. For instance, if accessing a lateral requires shutting and de-completing the entire well, the completion design may not be feasible for efficient production management. Conversely, completions that allow lateral access without well shutdown but lack real-time monitoring capabilities for pressure and temperature may limit reservoir management and early problem detection.

Therefore, a holistic approach to completion design is essential. This approach integrates junction selection, completion hardware, and monitoring capabilities to maximize the well's operational flexibility, production efficiency, and reservoir management. Intelligent completion systems that enable real-time flow control and pressure monitoring at each lateral further enhance the value of multilateral wells by allowing optimized production and early detection of issues.

Well Construction Design and Challenges

Window cutting is a critical aspect of well construction design, as it directly influences the choice of technology and the overall success of the multilateral well. Several key factors must be carefully considered during the design phase, including formation type, wellbore trajectory, and casing size, all of which impact the feasibility and efficiency of later operations.

When the window is cut in the build section of the wellbore, it typically occurs at a low angle. This positioning facilitates easier and more cost-effective lateral access during future interventions, as tools can be deployed with less complexity and risk. Although placing the window in this section may increase initial well construction costs, the benefits in terms of simplified and more reliable intervention operations often justify the investment. In such cases, TAML Level 4 technology is generally preferred, as it enables the junction and lateral to be fully cased and cemented, enhancing well integrity and mechanical stability.

Conversely, if the window is positioned in the horizontal section of the well, the operational complexity during lateral access increases significantly. Accessing laterals in this configuration can be more challenging and costly, requiring specialized tools and techniques to navigate the geometry. Moreover, the casing size plays a crucial role in determining the diameter of the lateral window and the tubing that can be installed. This, in turn, limits the maximum size of intervention tools that can be deployed, potentially restricting the range of maintenance and stimulation operations.

In scenarios where the window is cut in the horizontal section and casing size constraints exist, the design often defaults to TAML Level 2. This level leaves the junction and lateral as open hole, uncased, and uncemented, which simplifies the construction but compromises well integrity and hydraulic isolation. While this approach reduces upfront costs, it may increase the risk of wellbore instability and limit intervention options over the life of the well.

Therefore, well construction design must target a careful balance between initial construction costs, long-term well integrity, and the operational flexibility required for lateral access and interventions. It is essential to evaluate not only the technical feasibility of window placement and junction design but also the types of tools and interventions anticipated throughout the well's lifecycle. A comprehensive design approach ensures that the well can be efficiently managed and maintained, maximizing production while minimizing costly workovers and downtime (Sharma, 2022).

Lateral Access Windows Design and Challenges

Lateral access windows (Fig-7) are a critical design element that significantly enhances the functionality and operational efficiency of multilateral well completions. These windows provide a direct pathway from the main bore into each drilled lateral, enabling a wide range of intervention activities-such as logging, stimulation, and production treatments-without the need for full wellbore retrieval or rig mobilization. This capability reduces operational downtime and improves well management flexibility.

The integration of lateral access windows facilitates seamless interventions in both vertical and horizontal laterals. This is particularly important for routine logging and monitoring operations, often conducted in the vertical section of the well, which acts as an observation lateral. By allowing intervention tools to be deployed through the mother bore and into the laterals, lateral access windows support timely operational decisions and efficient well maintenance, ultimately maximizing production and reservoir management (Sharma, 2022).

Figure 7—Lateral access window placed across the lateral

Main Components of Lateral Access Windows

The lateral access window is an integral component of the completion system. It is installed and retrieved together with the completion tubing and accurately aligned opposite to the lateral entry point. This window features profiles designed to support a range of accessories, allowing the deployment of coiled tubing or slickline tools into either the main bore or the lateral. Two essential accessories required for accessing both the lateral and main bore through the lateral re-entry access window are the pressure isolation sleeve and the tubing window exit whipstock.

Pressure Isolation Sleeve

Pressure isolation sleeve (Fig-8) is a tool designed to be installed within the lateral re-entry window to provide hydraulic isolation of the lateral section. It features seals at both the top and bottom to ensure effective sealing, while its internal diameter is sized to allow intervention tools such as coiled tubing or slickline to pass through freely. The dimensions of the isolation sleeve are tailored to the size of the lateral re-entry window for compatibility and sealing integrity. This sleeve is deployed and retrieved using specific running and retrieving tools, which can be operated via slickline or coiled tubing.

Figure 8—Isolation sleeve installed inside the lateral access window

Tubing Exit Whipstock

The tubing exit whipstock or exit deflector (Fig-9) is a tool designed for installation in the re-entry window, featuring a deflector that must be precisely oriented toward the lateral to enable smooth and reliable access. The exit deflector is run and retrieved using specialized running and retrieving tools and can be operated via coiled tubing or slickline, depending on the angle and position of the window.

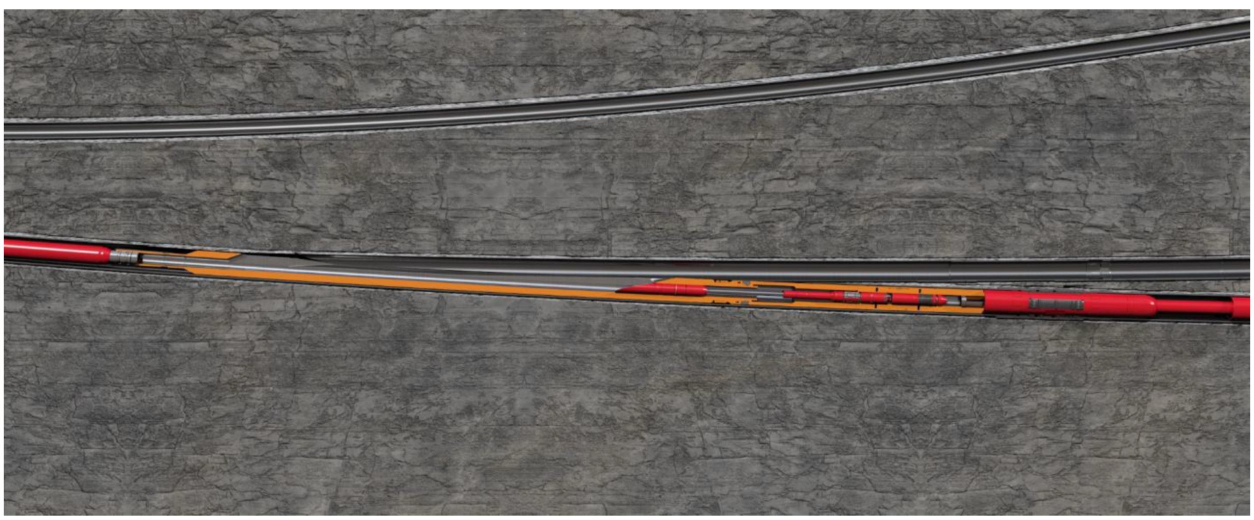

Figure 9—TEW installed inside the window to allow latera access

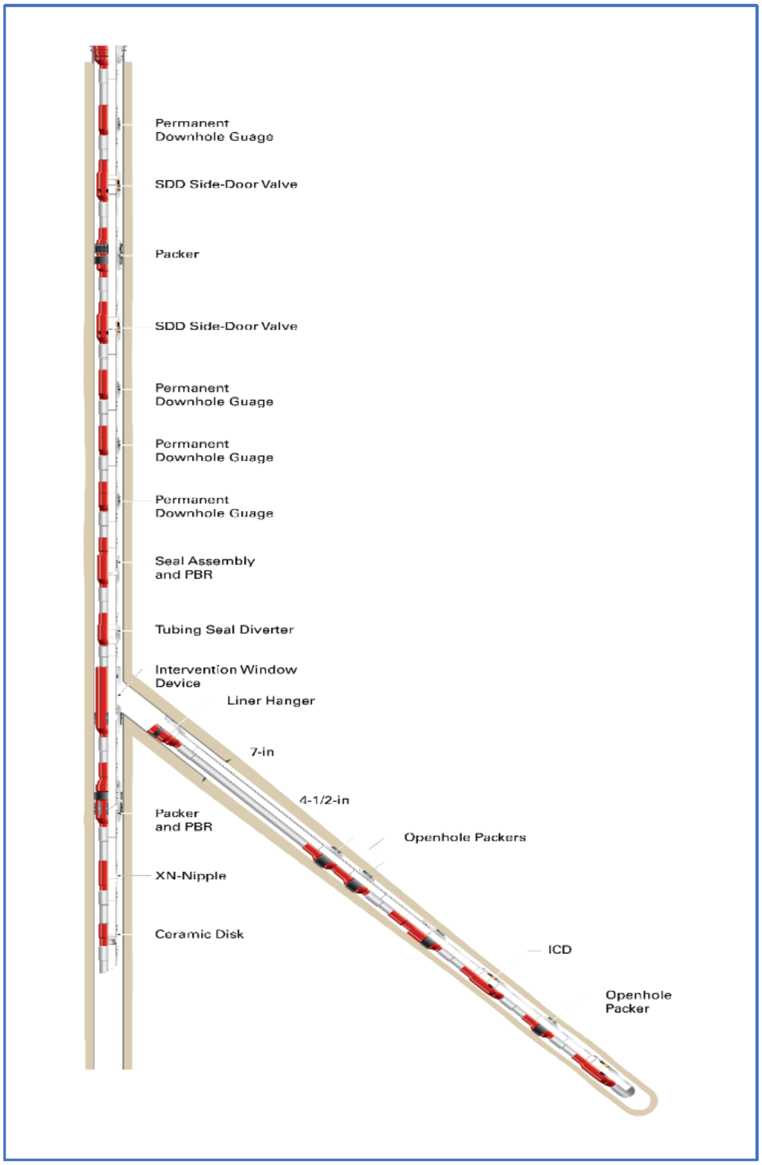

Figure 10—Schematic of Multilateral well showing the window installed the PDG

Design Challenges and Considerations

Designing lateral access windows involves several technical challenges. Precise orientation and placement are essential to ensure that intervention tools can be deployed efficiently throughout the well's life.

Latch couplings integrated into lateral access windows provide a permanent, full-bore internal diameter profile that allows tools to be repeatedly landed and oriented precisely at the window throughout the well's operational life. This feature is crucial for maintaining reliable access and orientation for interventions, ensuring consistent tool deployment and operational efficiency.

The size of the lateral window and associated components also impacts the maximum size of intervention tools that can be run, which influences the range of possible well treatments and maintenance operations. Additionally, ensuring pressure isolation and hydraulic integrity at the junction is vital to prevent cross-flow between laterals and maintain well control (Sharma, 2022).

Challenges Faced in Multilateral Well Window Placement and Intervention

The angle at which the lateral access window is installed is primarily determined by the location where the window is cut in the wellbore. Several factors influence the optimal placement of the junction, but the window's angle critically dictates the methodology used for later accessing the lateral during interventions.

- **Low-Angle or Deviated Windows:**
    - When the window is cut at a low angle or within the deviated section of the well, intervention tools such as slickline (SL) or slickline with tractor assistance can be effectively used to access the lateral. This approach tends to be less complex and more cost-efficient.
- **Horizontal Section Windows:**
    - If the window is cut in the horizontal section, the lateral re-entry window will share this orientation, making access more challenging. In this case, coiled tubing is typically required for intervention due to the increased difficulty in navigating horizontal laterals.

The decision on where to cut the window depends on multiple operational and geological factors. However, it is essential to carefully consider intervention frequency and complexity during the design phase, especially if frequent interventions are anticipated. Improper window placement can lead to increased operational costs and reduced intervention efficiency.

Another critical challenge is the size of the window, which is constrained by the casing size at the window location. The window size directly influences the maximum diameter of tools that can be deployed into both the lateral and the main bore. If this is not properly accounted for during well design, many intervention tools may become unusable, limiting the ability to perform essential reservoir management operations and potentially compromising production optimization.

Furthermore, well completion designs often require the integration of accessories such as permanent downhole gauges and additional nipple profiles. These components can complicate interventions by necessitating multiple trips into the wellbore, increasing operational time and cost (Sharida, 2025).

Challenges and Impact of Permanent Downhole Gauges in Multilateral Completions

Integrating permanent downhole gauges (PDGs) into multilateral well completions allow for continuous, real-time monitoring of pressure and temperature throughout the wellbore and reservoir. This steady flow of data is essential for maximizing production efficiency, identifying wellbore integrity issues early, and supporting proactive intervention planning, all of which contribute to enhanced well performance and safety. The presence of PDGs also increases the reliability of the overall system and enables intelligent completion strategies that can dynamically adapt to changing reservoir conditions. In multipurpose well designs, where the main bore functions as an observation well and the lateral as the producer, dedicated PDG monitoring systems for each section deliver unique and valuable insights. However, relying on a single PDG to monitor both sections is not sufficient; separate gauges must be installed and configured so that their readings do not interfere, ensuring the accuracy and integrity of the collected data.

Careful design of the well and completion system is therefore necessary to maintain independent monitoring streams for each section, supporting effective reservoir management and long-term well performance (Walaa, 2025).

Well Design and Access Methodology

Multilateral Well Design

For a multipurpose or dual producer well, the multilateral well design should incorporate a permanent downhole gauge (PDG) to enable continuous monitoring and data analysis. Additionally, a lateral access window should be installed along the lateral section to allow entry into both the main bore and the lateral.

In a multipurpose well, the main bore typically serves as an observation well for periodic logging and data collection, while the lateral functions as the primary producer. However, maintaining an open window between the main bore and the lateral can compromise the integrity of PDG readings. To address this, the well design must allow for isolation of the main bore from the lateral. This is achieved by installing an isolation sleeve within the lateral access window, which remains in place throughout the well's production life. As a result, production from the lateral is directed into the annulus and then into the production tubing via a side door sliding sleeve (SSD). For comprehensive monitoring, the well should be equipped with two PDGs: one to measure pressure and temperature in the annulus (reflecting the producing lateral), and another to monitor the main bore. To further enhance integrity, a plug is installed in a nipple positioned between the two PDGs (Fig-4).

Access Methodology

Main Bore Access

To access the main bore:

1. Retrieve the plug installed in the nipple between the two PDGs.
2. Run in hole with the desired logging tool.

Lateral Access

To access the lateral:

1. Retrieve the plug installed between the two PDGs.
2. Retrieve the isolation sleeve from the lateral access window.
3. Install an exit whipstock to guide the logging tool into the lateral.
4. Run in hole with the logging tool.

Transformative Value to The Intervention of Multilateral

Multilateral wells have gained significant importance in the oil and gas sector, even though intervention challenges have sometimes limited their broader adoption. However, recent innovations in well design and technology have transformed their potential. These new approaches ensure reliable, long-term access to both the lateral and main bore sections, making it much easier to manage and maintain the well throughout its lifecycle. The ability to install dedicated permanent downhole gauges (PDGs) for each lateral—without interference between them adds even greater value. This design gives the industry real-time data and analytics, empowering the industry to plan and execute interventions more effectively and efficiently.

Conclusion

Multilateral well technology has established itself as a transformative solution for optimizing field development in complex and mature reservoirs. By enabling multiple lateral branches from a single main bore, these wells maximize reservoir contact, increase hydrocarbon recovery, and minimize environmental footprint and operational costs. The integration of advanced completion technologies including lateral access windows, and permanent downhole gauges (PDGs) ensures reliable, long-term access for interventions and continuous, real-time monitoring of well performance.

The design of multilateral wells must carefully balance technical feasibility, well integrity, and operational flexibility. Selection of appropriate junction types, precise placement of lateral access windows, and the use of isolation sleeves and PDGs are critical to maintaining data integrity and supporting effective reservoir management. These design choices enable the industry to address intervention challenges, optimize production, and adapt to changing reservoir conditions throughout the well's lifecycle.

Recent innovations in multilateral well design have significantly improved reliability and performance, making these wells a practical and forward-looking choice for modern field development. The ability to access, monitor, and manage both the main bore and laterals independently while minimizing interference and maximizing data accuracy delivers substantial value to the oil and gas sector. As technology continues to advance, multilateral wells will play an increasingly vital role in unlocking the full potential of complex reservoirs and supporting sustainable, efficient hydrocarbon production.

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